

CHINMOY BHATTACHARJEE

JUNIOR PROFESSOR OF MATHEMATICAL STOCHASTICS · Universität Hamburg

PERSONAL DETAILS

PLACE | DATE OF BIRTH: Kolkata, India | 6 December 1990
NATIONALITY: Indian
OFFICE ADDRESS: Department of Mathematics, Universität Hamburg
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ACADEMIC POSITIONS

SEP. 2026–present	Junior Professor (W1) at Department of Mathematics , University of Hamburg
JUN. 2026	Positive interim evaluation (Habilitationsäquivalent) Extension of Junior Professorship until AUG. 2029
SEP. 2023–AUG. 2026	Junior Professor (W1) [first phase] at Department of Mathematics , University of Hamburg
AUG. 2020–AUG. 2023	Postdoctoral Researcher at Department of Mathematics , University of Luxembourg
AUG. 2018–JUL. 2020	Postdoctoral Researcher at Institut für Mathematische Statistik und Versicherungslehre (IMSV), University of Bern
JUL. 2018–AUG. 2018	Visiting Scientist at Indian Statistical Institute , Stat-Math Unit, Kolkata

EDUCATION

AUG. 2013–MAY 2018	PhD in Applied Mathematics , Department of Mathematics , University of Southern California ADVISOR: Prof. Larry Goldstein THESIS: Stein's method and its applications in strong embeddings and Dickman approximations
AUG. 2011–MAY 2013	Master of Statistics (M. Stat.) , Indian Statistical Institute , Kolkata SPECIALIZATION: Mathematical Statistics and Probability
AUG. 2008–MAY 2011	Bachelor of Statistics with Honours (B. Stat. (Hons.)) , Indian Statistical Institute , Kolkata

PREPRINTS

1. **C. Bhattacharjee**, A. Gusakova, and C. Hirsch. Functional central limit theorems for Laguerre tessellations. Preprint, 2026+. [arXiv: 2610.06372](#)
2. **C. Bhattacharjee**. Dickman approximation of randomly weighted sums via Stein's method. Submitted to *Annales Henri Lebesgue*, 2026+. [arXiv: 2606.27875](#)
3. **C. Bhattacharjee**, and A. Gusakova. Gaussian approximation for extreme points in Laguerre tessellations. Submitted to *Electronic Journal of Probability*, 2025+. [arXiv: 2510.21665](#)
4. **C. Bhattacharjee**, G. Peccati, and D. Yogeshwaran. Spectra of Poisson functionals and applications to continuum percolation. Submitted to *Annals of Applied Probability*, 2024+. [arXiv: 2407.13502](#)
5. Z. Shi, **C. Bhattacharjee**, K. Balasubramanian, and W. Polonik. Multivariate Gaussian approximation for random forest via region-based stabilization. Submitted to *Annals of Applied Probability*, 2024+. [arXiv: 2403.09960](#)
6. **C. Bhattacharjee**, and R. O'Clarey. Chaos and superconcentration for Poisson functionals with applications in stochastic geometry. Major revision in *Electronic Journal of Probability*, 2026+. [arXiv: 2603.23053](#)

PEER-REVIEWED PUBLICATIONS

7. **C. Bhattacharjee**, R. Versendaal, and A. Wade. Perimeter length of the convex hull of Brownian motion in the hyperbolic plane. *Journal of Applied Probability*, published online, 2026.
8. Z. Shi, **C. Bhattacharjee**, K. Balasubramanian, and W. Polonik. Gaussian and bootstrap approximation for matching-based average treatment effect estimators. *Annals of Statistics*, 54(4), 2164–2190, 2026.
9. **C. Bhattacharjee**, and M. Schulte. Dickman approximation of weighted sums of independent random variables in the Kolmogorov distance. *Annals of Applied Probability*, 35(5), 3271–3309, 2025.
10. **C. Bhattacharjee**, I. Molchanov, and R. Turin. Central limit theorem for a birth–growth model with Poisson arrivals and random growth speed. *Advances in Applied Probability*, 56(3), 1004–1032, 2024.
11. **C. Bhattacharjee**, and I. Molchanov. Gaussian approximation for sums of region-stabilizing scores. *Electronic Journal of Probability*, 27, paper no. 111, 2022.
12. **C. Bhattacharjee**. Gaussian approximation for rooted edges in a random minimal directed spanning tree. *Random Structures & Algorithms*, 61(3), 462–492, 2022.
13. **C. Bhattacharjee**, and M. Schulte. Large degrees in scale-free inhomogeneous random graphs. *Annals of Applied Probability*, 32(1), 696–720, 2022.
14. **C. Bhattacharjee**, and I. Molchanov. Convergence to scale-invariant Poisson processes and applications in Dickman approximation. *Electronic Journal of Probability*, 25, paper no. 79, 2020.

15. **C. Bhattacharjee**, and L. Goldstein. Dickman approximation in simulation, summations and perpetuities. *Bernoulli*, 25(4A), 2758–2792, 2019.
16. **C. Bhattacharjee**, and L. Goldstein. On strong embeddings by Stein’s method. *Electronic Journal of Probability*, 21, paper no. 15, 2016.

GRANTS

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| 2024–2027
(36 MONTHS) | Superconcentration in random geometric structures
DFG Priority Program ‘Random Geometric Systems’ (SPP 2265)
ROLE: Principal Investigator
AMOUNT: 219,966 Euros |
| 2026–2028
(24 MONTHS) | Sharp noise sensitivity & exceptional times in continuum percolation
Programmes for Project-Related Personal Exchange (PPP) [DST & DAAD]
ROLE: co-PI, together with Prof. D. Yogeshwaran (ISI, Bangalore, India)
AMOUNT: 11,308 Euros |

INVITED LECTURES (IN LAST FIVE YEARS)

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| DEC. 2026 | Workshop on Stein’s Method and Its Statistical Applications , ESSEC Business School, Cergy, France (upcoming) |
| AUG. 2026 | European Meeting of Statisticians (EMS) , invited session <i>Stein-based inference</i> , Università della Svizzera italiana, Lugano, Switzerland |
| JUL. 2026 | Second Workshop on Stein’s Method and Networks , University of Oxford, UK |
| NOV. 2025 | Mathematics Colloquium , Institut für Mathematik, Universität Osnabrück, Germany |
| OCT. 2025 | Workshop on Random Geometry, Graphs and Extremes , IUC Dubrovnik, Croatia |
| APR. 2025 | Stochastics Seminar , Institut for Matematik, Aarhus Universitet, Denmark |
| OCT. 2024 | Workshop on Random Geometry and Particle Systems , IMT Nord Europe, Lille, France |
| AUG. 2024 | Bernoulli–IMS World Congress in Probability and Statistics , invited session <i>Stein’s method and random graphs</i> , Ruhr-Universität Bochum, Germany |
| JUN. 2024 | Workshop on Stein’s Method and Networks , University of Oxford, UK |
| JUN. 2024 | Workshop on Random Geometric Systems (SPP 2265), Technische Universität Dortmund, Germany |
| APR. 2024 | Stochastics Seminar , Institut für Mathematische Stochastik, Universität Münster, Germany |
| SEP. 2022 | Mini-Workshop: Limit Theorems for Spatial Random Structures , Ruhr-Universität Bochum, Germany |
| JUN. 2022 | Stein’s Method: The Golden Anniversary , Institute for Mathematical Sciences, National University of Singapore |

TEACHING EXPERIENCE

UNIVERSITY OF HAMBURG	Stochastische Prozesse (upcoming, in German) Winter 2026/27
	Stochastik 1 für Studierende der Informatik (in German) Summer 2025, Summer 2026 (jointly with Prof. Dr. Johannes Lederer)
	Seminar on Probabilistic Methods in Combinatorics Summer 2025, Summer 2026
	Stochastik 2 für Studierende der Informatik (in German) Winter 2025/26 (jointly with Prof. Dr. Johannes Lederer)
	Seminar on Percolation theory Winter 2025/26
	Introduction to Random Graphs Master in Mathematics; Winter 2023/24, Winter 2024/25
	Introduction to Stein's Method Master in Mathematics; Summer 2024
UNIVERSITY OF LUXEMBOURG	Probability and Statistics II BMATH fourth semester; Spring 2023, Spring 2022
	Refresher in Probability Master of Data Science first semester; Fall 2022, Fall 2021
	Probabilistic Methods in Combinatorics BMATH fifth semester; Fall 2021, Fall 2020
UNIVERSITY OF BERN	Introduction to Random Graphs MMATH second semester; Spring 2019
UNIVERSITY OF SOUTHERN CALIFORNIA (ASSISTANCE)	Math 125 - Calculus I ; Fall 2017
	Math 308 - Statistical Inference and Data Analysis II ; Spring 2016
	Math 307 - Statistical Inference and Data Analysis I ; Fall 2015
	Math 226 - Calculus III ; Fall 2014, Spring 2015

SUPERVISION

PHD.	Rowan O'Clarey (from Oct. 2024 – Sep. 2027 Expected) -Funded by DFG grant no. 531540467 TOPIC: Superconcentration in random geometric structures
OTHER SUPERVISION	Amélie Boussemart (Master's thesis) Oliver Gertke (Master's thesis) Maika Lu Beecken (Bachelor's thesis)

ORGANIZATION & ACADEMIC SERVICE

Workshop Organization:

- MAR. 2025 **Session Co-chair:** *Spatial stochastics, disordered media, and complex networks*, in the *German Probability and Statistics Days*, Dresden, Germany.
Co-chaired with B. Jahnel (TU Braunschweig/WIAS Berlin)
- SEP. 2024 **Workshop Coorganizer:** *Workshop on Stein's method and stochastic geometry*, Hamburg, Germany.
Coorganized with M. Schulte (TUHH, Hamburg)

Peer Reviewing:

- JOURNALS Annals of Statistics (AoS), Applied Probability Journals, Bernoulli, Extremes, Electronic Journal of Probability (EJP), Journal of the American Statistical Association (JASA), Probability Theory & Related Fields (PTRF), Random Structures & Algorithms (RSA), Statistics & Probability Letters, Stochastic Processes & their Applications (SPA)

LANGUAGE SKILLS

Bengali (mother tongue), English (C2), German (B2), French (A1), Hindi (C1)